PEN PALS

Mth632 - Quiz 3 solved 2021

Question # 7 of 10 (Start time: 08:13:38 PM, 13 August 2021)

Let C be simple closed contour in simple connected domain D and f be an analytic function in D and z_o be a point that lies in C then $f(z_o)$ is

Select the correct option

- $\int_C \frac{f(z)}{(z-z_o)}dz$
- $\frac{1}{\pi i}\int_C \frac{f(z)}{(z-z_o)}dz$
- $\frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_o)}dz$ ✓
- $2\pi i\int_C \frac{f(z)}{(z-z_o)}dz$

Click to Save An

Consider the integral $\int_C f(z)dz$ where contour C:$z(t) = e^{it}$, $L = 2\pi$ and $f(z) = \bar{z}$. The upper bound is

ect the correct option

- π
- 2π
- -2π
- $-\pi$

Question # 5 of 10 (Start time: 08:10:59 PM. 13 August 2021)

If a function $f(z) = e^{iz}$ *and its derivatives are analytic,*

and C *is simple closed contour that bounds some domain which contain* z_o, *then* $\int_C \frac{f(z)}{(z - z_o)^4} dz$

Select the correct option

- $\frac{2\pi}{3} e^{iz_o}$
- $\frac{2\pi i}{3} e^{iz_o}$ ✓
- $\frac{\pi}{3} e^{iz_o}$
- $\frac{\pi i}{3} e^{iz_o}$

Question # 4 of 10 (Start time: 08:09:50 PM, 13 August 2021)

$Consider\ contour\ C : z(t) = e^{it} = Cos\ t\ + i\ Sint\ be\ counter\ clockwise\ where\ -\pi/2 \leqslant t \leqslant \pi/2\ then\ the\ integral$

$$\int_C \bar{z}(t)dz =$$

Select the correct option

- $-i\pi$
- $-\pi$
- $i\pi$
- π

Question # 7 of 10 (Start time: 08:05:42 PM. 13 August 2021)

If function f is an analytic function in simple connected domain where $C_R(z_o) = \{z(t) : |z - z_o| = R\}, 0 \leqslant t \leqslant 2\pi$ and R be the radius of circle, then

Select the correct option

Reload

- $f(z_o) = \frac{1}{2\pi i} \int_0^{2\pi} f(z_o + \mathrm{Re}^{it}) dt$
- $f(z_o) = \frac{1}{2\pi} \int_0^{2\pi} f(z_o + \mathrm{Re}^{it}) dt$
- $f(z_o) = \frac{1}{\pi} \int_0^{2\pi} f(z_o + \mathrm{Re}^{it}) dt$
- $f(z_o) = \frac{1}{4\pi} \int_0^{2\pi} f(z_o + \mathrm{Re}^{it}) dt$

Click to Save Answer & Move to N

Question # 6 of 10 (Start time: 08:04:45 PM, 13 August 2021)

Morera's Theorem is the converse of ...

Select the correct option

- Guass Mean value Theorem
- De Movire's theorem
- Cauchy Goursat Theorem
- Green's Theorem

MTH632:Quiz # 3

Question # 4 of 10 (Start time: 08:03:47 PM, 13 August 2021)

A domain D that is not simply connected is...

Select the correct option

- ○ none of above
- ● multiply connected
- ○ triply connected
- ○ doubly connected

If C is simple closed contour with positively oriented such that fixed complex value z_o lies interior to C, then for integer $n \neq 1$ we have,

$$\int_C \frac{1}{(z - z_o)^n} dz =$$

Select the correct option

- 0
- πi
- $2\pi i$
- $-2\pi i$

Question # 5 of 10 (Start time: 07:55:26 PM, 15 August 2021)

If the interior of every simple closed contour C is contained in the domain D , then domain is said to be....

Select the correct option

- None of above
- Simple connected
- Unbounded
- Disconnected

Question # 8 of 10 (Start time: 07:58:14 PM, 13 August 2021)

Evaluate $\int_C f(z)dz$ where C:$f(z) = e^{it}$ for $0 \leqslant t \leqslant 2\pi$ and $f(z) = 1/z$

Select the correct option

- $2\pi i$
- πi
- $-2\pi i$
- $-\pi i$

Question # 7 of 10 (Start time: 07:57:00 PM, 13 August 2021) **Total Mark**

```
\begin{gathered}
  {\text{Let f(z)\;be analytic function in simply connected domain D and }}{{\text{f}}^{'}}^{\text{\;}}{\text{is continuous in D}}{\text{. }} \hfill \\  {\text{If C is closed contour lies in D then }}\int\limits_{\text{C}} {{\text{f(z) dz}}} {\text{\; = 0}}\; \hfill \\
\end{gathered}
```

Select the correct option

Reload Math Equatio

- ○ Cauchy Theorem
- ○ Rolle's Theorem
- ○ Green's Theorem
- ○ Mean Value Theorem

Question # 6 of 10 (Start time: 07:56:09 PM, 13 August 2021)

If $f(z)$ is an analytic function in domain D then for $n \geqslant 0$ where n is a positive integer,

Select the correct option

○	None of above
○	$f^{(n)}(z)$ is not analytic in simple connected D
○	$f^{(n)}(z)$ is analytic in disconnected D
○	$f^{(n)}(z)$ is an analytic in simple connected D



Click to Save

Question # 2 of 10 (Start time: 07:52:22 PM, 13 August 2021)

If $f(z)$ is complex valued function and C and $-C$ be the contours with same geometrical representation but opposite in direction. Then $\int_{-C} f(z)dz =$

Select the correct option

Reload Ma

- $-\int_{C} f(z)dz$
- $\int_{C} f(-z)dz$
- $-\int_{-C} f(z)dz$
- $\int_{C} f(z)dz$

Question # 4 of 10 (Start time: 07:54:01 PM, 13 August 2021)

A continuous complex function f(z) defined on a set containing contour C and z(t) be any parameterization of C for a ≤ t ≤ b then

$$\int_C f(z)dz =$$

Select the correct option

- ○ $\int_a^b f(z)z(t)dz$
- ○ \int\limits_a^b {{f^'}(z)z(t)dz}
- ○ \int\limits_a^b {{f^'}(z){z^'}(t)dz}
- ○ \int\limits_a^b {f(z){z^'}(t)dz}

Question # 5 of 10 (Start time: 07:55:33 PM, 13 August 2021)

If domain D is simply connected, then any contour of integral around simple closed curve is

Select the correct option

- -1
- 0
- 2
- 1

MTH632:Quiz # 3

Question # 1 of 10 (Start time: 07:51:31 PM, 13 August 2021)

Let C_1 and C_2 be two simple closed contours with positively oriented such that $C_1 - C_2$ is positively oriented as boundary of the region R then

Select the correct option

- $\int_{C_1 - C_2} f(z)dz \geqslant 0$
- $\int_{C_1 - C_2} f(z)dz = 0$
- $\int_{C_1 - C_2} f(z)dz \neq 0$
- $\int_{C_1 - C_2} f(z)dz \leqslant 0$

Question # 10 of 10 (Start time: 07:45:39 PM, 13 August 2021)

If C is simple closed contour and D be the domain that forms interior of C. P and Q are continuous function and their partial derivatives are also continuous then

Select the correct option

- $\int_C [P(x,y)dx+Q(x,y)dy] = \iint_D (Q_x(x,y) - P_y(x,y))dxdy$
- $\int_C [P(x,y)dx-Q(x,y)dy] = \iint_D (Q_x(x,y) + P_y(x,y))dxdy$
- $\int_C [P(x,y)dx+Q(x,y)dy] = \iint_D (Q_x(x,y) + P_y(x,y))dxdy$
- $\int_C [P(x,y)dx-Q(x,y)dy] = \iint_D (Q_x(x,y) - P_y(x,y))dxdy$

Question # 8 of 10 (Start time: 07:42:49 PM, 13 August 2021)

Let $f(t)$ be continuous function on contour C and L be length of C. For all t in contour C, $|f(t)| \leqslant M$ then

Select the correct option

- $\left| \int_C f(t)\, dt \right| > ML$
- $\left| \int_C f(t)\, dt \right| < ML$
- $\left| \int_C f(t)\, dt \right| \leqslant ML$ ✓
- $\left| \int_C f(t)\, dt \right| \geqslant ML$

Question # 5 of 10 (Start time: 07:39:51 PM. 13 August 2021)

Let α lies in the int *erior of contour C and*

$$P_n(z) = a_n z^n + a_{n-1} z^{n-1} + a_{n-2} z^{n-2} + \ldots + a_1 z + a_o z^0$$

then the value of $\int_C \frac{P_n(z)}{z-\alpha} dz$ *is ... when* $n = 3$

Select the correct option

- $2\pi i P_3(\alpha)$
- $2\pi P_n(\alpha)$
- $2\pi P_3(\alpha)$
- $2\pi i P_n(\alpha)$

MTH632:Quiz # 3

Question # 6 of 10 (Start time: 07:40:49 PM, 13 August 2021)

Evaluate the integral $\int_C \frac{e^z}{z^4} dz$ where the contour $C: \ |z| = 1$

Select the correct option

○	$\frac{\pi i}{3}$
○	$\frac{2\pi i}{3}$
○	$\frac{4\pi i}{3}$
○	$\frac{8\pi i}{3}$

Question # 7 of 10 (Start time: 07:41:59 PM, 13 August 2021)

$$Evaluate \int_C f(z)dz \; where \; C : z(t) = (x + iy)\, t \; for \; a \leqslant t \leqslant b \; and f(z) = z$$

Select the correct option

○	$\frac{(b^2 - a^2)(x + iy)}{2}$
○	$\frac{(b^2 - a^2)(x + iy)^2}{2}$
○	$\frac{(b^2 + a^2)(x + iy)^2}{2}$
○	$\frac{(b^2 + a^2)(x + iy)}{2}$

Question # 4 of 10 (Start time: 07:39:19 PM, 13 August 2021)

A curve is said to be .... if it is constructed by joining finitely many smooth curves end to end

Select the correct option

○	disconnected
○	discontinuous
○	contour
○	continuous

Question # 3 of 10 (Start time: 07:38:23 PM, 13 August 2021)

A domain D (connected open set) is called _____ connected if interior of every simple closed contour C is contained in D.

Select the correct option

- simply
- multiply

MTH632:Quiz # 3

Quiz Start Time: 08:25 PM, 13 August 2021

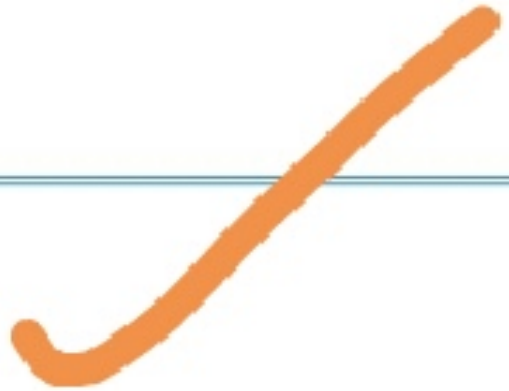
Question # 10 of 10 (Start time: 08:31:42 PM, 13 August 2021)

Total Marks: 1

Each simple closed curve C divides the plane into two domains (connected open sets). One of which is bounded and is called _____ of C.

Select the correct option

- exterior
- interior



Click to Save Answer & Move to Next Question

MTH632:Quiz # 3

Quiz Start Time: 08:25 PM, 13 August 2021

Question # 4 of 10 (Start time: 08:27:54 PM, 13 August 2021)

Total Marks: 1

Let $f(z)$ be an analytic in a domain D and D be the interior and boundary of th circle 'r' defined as $|z-a|=r$ and $|f(z)| \leqslant M$, then $|f^{(n)}(a)| \leqslant \frac{n!M}{r^n}$

Select the correct option

Reload Math Equations

- ML-Inequality
- None of above
- Cauchy Inequality
- Cauchy Sichwarz Inequality

MTH632:Quiz # 3

Quiz Start Time: 08:25 PM, 13 August 2021

Question # 7 of 10 (Start time: 08:29:31 PM, 13 August 2021)

Total Marks: 1

Let z_o and z_1 be two points in simply connected domain D and f be an analytic complex valued function in D and C be contour by joining z_o and z_1 then

Select the correct option

Reload Math Equations

- $\int_{z_o}^{z_1} f(s)ds = F(z_1) - F(z_o)$
- $\int_{z_o}^{z_1} f(s)ds = F(z_o) - F(z_1)$
- $\int_{z_o}^{z_1} f(s)ds = F(z_1) + F(z_o)$
- $\int_{z_o}^{z_1} f(s)ds = F(z_o) + F(z_o)$

Click to Save Answer & Move to Next Question

sec(s)

MTH632:Quiz # 3

Quiz Start Time: 08:25 PM, 13 August 2021

Question # 9 of 10 (Start time: 08:30:49 PM, 13 August 2021)

Total Marks: 1

If the function f(z) is analytic in domain D and simple connected contour C lies in D then

Select the correct option

Reload Math Equations

- $\int_C f(z)dz = 0$
- $\int_C f(\bar{z})dz = 0$
- $\int_C f(z')dz = 0$
- $\int_C f(-z)dz = 0$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 08:09:50 PM, 13 August 2021)

$Consider\ contour\ C : z(t) = e^{it} = Cos\ t\ + i\ Sint\ be\ counter\ clockwise\ where\ -\pi/2 \leqslant t \leqslant \pi/2\ then\ the\ integral$

$$\int_C \bar{z}(t)dz =$$

Select the correct option

- $-i\pi$
- $-\pi$
- $i\pi$
- π

Question # 5 of 10 (Start time: 08:10:59 PM. 13 August 2021)

If a function $f(z) = e^{iz}$ *and its derivatives are analytic,*

and C *is simple closed contour that bounds some domain which contain* z_o, *then* $\int_C \frac{f(z)}{(z - z_o)^4} dz$

Select the correct option

- $\frac{2\pi}{3} e^{iz_o}$
- $\frac{2\pi i}{3} e^{iz_o}$ ✓
- $\frac{\pi}{3} e^{iz_o}$
- $\frac{\pi i}{3} e^{iz_o}$

Consider the integral $\int_C f(z)dz$ where contour C:$z(t) = e^{it}$, $L = 2\pi$ and $f(z) = \bar{z}$. The upper bound is

ect the correct option

- π
- 2π
- -2π
- $-\pi$

Question # 9 of 10 (Start time: 08:16:03 PM, 13 August 2021) Total Marks: 1

Consider function $f(z) = \exp(z^2 + 1)$ be differentiable and its derivatives are continuous in domain D then $\int_C f(z)dz = \ldots\ldots$ where C is simple closed contour in domain D

Select the correct option

Reload Math Equations

- 0
- -1
- 2
- 1

Question # 8 of 10 (Start time: 08:14:49 PM, 13 August 2021)

$$Evaluate\ the\ integral\ \int_C f(z)dz\ \ where\ C: z(t) = e^{it}\ for\ 0 \leqslant t \leqslant \pi\ and f(z) = 1/z^2$$

Select the correct option

○	$1 - e^{-i\pi}$
○	$1 - e^{i\pi}$
○	$1 + e^{i\pi}$
○	$1 + e^{-i\pi}$

Question # 7 of 10 (Start time: 08:13:38 PM, 13 August 2021)

Let C be simple closed contour in simple connected domain D and f be an analytic function in D and z_o be a point that lies in C then $f(z_o)$ is

Select the correct option

- $\int_C \frac{f(z)}{(z-z_o)}dz$
- $\frac{1}{\pi i}\int_C \frac{f(z)}{(z-z_o)}dz$
- $\frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_o)}dz$
- $2\pi i\int_C \frac{f(z)}{(z-z_o)}dz$

Click to Save An

Question # 10 of 10 (Start time: 08:16:37 PM, 13 August 2021)

Evaluate the integral $\int_C \frac{e^z}{z^4} dz$ where the contour $C: \ |z| = 1$

Select the correct option

○	$\frac{\pi i}{3}$
○	$\frac{2\pi i}{3}$
○	$\frac{8\pi i}{3}$
○	$\frac{4\pi i}{3}$

Question # 5 of 10 (Start time: 05:10:23 PM, 13 August 2021) Total Marks: 1

Let z_o and z_1 be two points in simply connected domain D
and f be an analytic complex valued function in D and C be contour by joining z_o and z_1 then

Select the correct option

Reload Math Equations

- $\int_{z_0}^{z_1} f(s)ds = F(z_1) + F(z_o)$
- $\int_{z_0}^{z_1} f(s)ds = F(z_o) + F(z_o)$
- $\int_{z_0}^{z_1} f(s)ds = F(z_o) - F(z_1)$
- $\int_{z_0}^{z_1} f(s)ds = F(z_1) - F(z_o)$

Click to Save Answer & Move to Next Question

Evaluate $\frac{1}{2\pi}\int_0^{2\pi} \exp(\frac{i\pi}{3} + 5e^{i\theta})d\theta$ if $f(z) = \exp(z)$ then we get,

Select the correct option

Reload Math Equations

- $\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$
- $\sin\frac{\pi}{3} + i\cos\frac{\pi}{3}$
- $\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}$
- $\cos\frac{\pi}{3} + \sin\frac{\pi}{3}$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 05:09:03 PM, 13 August 2021) Total Marks: 1

Using Cauchy integral formula to compute the value of integral

$$\int_C \frac{z-2}{z+i} dz \text{ where } f(z) = z-2 \text{ and } z_o = -i$$

Select the correct option

Reload Math Equations

- $2\pi(1-2i)$
- $2\pi(-1+2i)$
- $2\pi(-1-2i)$ ✓
- $2\pi(1+2i)$

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 05:00:12 PM, 13 August 2021)

Let C_1 and C_2 be two simple closed contours with positively oriented s. t C_1 is exterior and C_2 is an interior. If $f(z)$ is analytic in domain

Select the correct option

- $\int_{C_1} f(z)dz = \int_{C_2} f(z)dz$
- $\int_{C_1} f(z)dz > \int_{C_2} f(z)dz$
- $\int_{C_1} f(z)dz \neq \int_{C_2} f(z)dz$
- $\int_{C_1} f(z)dz < \int_{C_2} f(z)dz$

f 10 (Start time: 05:09:12 PM, 13 August 2021)

C_2 be two simple closed contours with positively oriented s.t C_1 is exterior and C_2 is an interior. If $f(z)$ is analytic in domai

ect option



$$\int_{C_1} f(z)dz = \int_{C_2} f(z)dz$$

$$\int_{C_1} f(z)dz > \int_{C_2} f(z)dz$$

$$\int_{C_1} f(z)dz \neq \int_{C_2} f(z)dz$$

$$\int_{C_1} f(z)dz < \int_{C_2} f(z)dz$$

Time Left

MTH632:Quiz # 3

Quiz Start Time: 04:50 PM, 13 August 2021

Question # 10 of 10 (Start time: 04:57:35 PM, 13 August 2021)

Total Marks: 1

$$Evaluate \quad \int_C f(z)dz \quad where\ C : z(t) = e^{it}\ for\ 0 \leq t \leq \pi\ and\ f(z) = 1/z$$

Select the correct option

Reload Math Equations

- $i\pi$
- $-2\pi i$
- $2\pi i$
- $-\pi i$

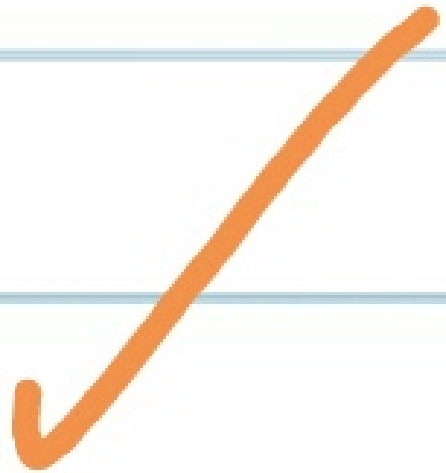
Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 04:58:14 PM, 13 August 2021) Tot

Each simple closed curve C divides the plane into interio and exterior domains where one is bounded and other is unbounded. This statement is of .....

Select the correct option

- Green's Theorem
- Cauchy Theorem
- Cauchy Goursat Theorem
- Jordan Curve Theorem



tegral formula to compute the value of integral

$$\int_C \frac{z-2}{z+i} dz \text{ where } f(z) = z-2 \text{ and } z_0 = -i$$

option

Reload Math

$2\pi(-1-2i)$

$2\pi(-1+2i)$

$2\pi(1+2i)$

$2\pi(1-2i)$

Click to Save Answer & Move to Next Questi

Question # 9 of 10 (Start time: 04:56:23 PM, 13 August 2021)

Total Marks: 1

Each simple closed curve C divides the plane into two domains (connected open sets). One of which is bounded and is called _____ of C.

Select the correct option

- interior
- exterior



Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 04:55:37 PM, 13 August 2021) Total Marks: 1

A curve is said to be .... if it is constructed by joining finitely many smooth curves end to end

Select the correct option

- disconnected
- contour
- continuous
- discontinuous



Question # 7 of 10 (Start time: 04:54:55 PM, 13 August 2021) Total Marks: 1

Mathematically, the functions in Green's theorem will be

Select the correct option

- Continuous derivatives
- Discrete derivatives
- Discrete partial derivatives
- Continuous partial derivatives



MTH632:Quiz # 3

Quiz Start Time: 04:50 PM, 13 August 2021

Question # 4 of 10 (Start time: 04:52:17 PM, 13 August 2021)

Total Marks: 1

$$\int_C \frac{f(z)}{(z+1)^4} dz \text{ where } z = e^{2z} \text{ then this integral is equal to}$$

Select the correct option

Reload Math Equations

- $\frac{\pi i}{3!} f^{(3)}(1)$
- $\frac{2\pi i}{3!} f^{(3)}(1)$ ✓
- $\frac{\pi i}{3!} f^{(3)}(-1)$
- $\frac{2\pi i}{3!} f^{(3)}(-1)$ ✓

Question # 5 of 10 (Start time: 04:52:55 PM, 13 August 2021) Total Marks: 1

Evaluate the integral $\int_C f(z)dz$ where $C: z(t) = e^{it}$ for $0 \leq t \leq \pi$ and $f(z) = 1/z^2$

Select the correct option

Reload Math Equations

- $1 - e^{-i\pi}$
- $1 + e^{-i\pi}$
- $1 - e^{i\pi}$
- $1 + e^{i\pi}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 04:53:27 PM, 13 August 2021) Total Marks: 1

Using Cauchy integral formula to compute the value of integral

$$\int_C \frac{z-2}{z+i} dz \text{ where } f(z) = z-2 \text{ and } z_a = -i$$

Select the correct option

Reload Math Equations

- $2\pi(-1+2i)$
- $2\pi(1+2i)$
- $2\pi(-1-2i)$ ✓
- $2\pi(1-2i)$

Click to Save and Move to Next Question

If the interior of every simple closed contour C is contained in the domain D , then domain is said to be....

Select the correct option

- None of above
- Unbounded
- Disconnected
- Simple connected ✓

MTH632:Quiz # 3

Quiz Start Time: 04:50 PM, 13 August 2021

Question # 2 of 10 (Start time: 04:50:53 PM, 13 August 2021)

Total Marks: 1

If C is simple closed contour with positively oriented such that fixed complex value z_0 lies interior to C, then for integer $n \neq 1$ we have,

$$\int_C \frac{1}{(z - z_0)^n} dz =$$

Select the correct option

Reload Math Equations

- 0
- πi
- $-2\pi i$
- $2\pi i$ ✓

Click to Save Answer & Move to Next Question

MTH632:Quiz # 3

Quiz Start Time: 04:50 PM, 13 August 2021

Question # 1 of 10 (Start time: 04:50:03 PM, 13 August 2021)

Total Marks: 1

If function f is continuous in simple connected domain D and $\int_C f(z)dz$ for every closed contour C in D, then f is analytic in D.

This statement is of:

Select the correct option

Reload Math Equations

- Green's Theorem
- Morera's Theorem
- Cauchy Goursat Theorem
- Cauchy theorem

sec(s)

MTH632:Quiz # 3

Quiz Start Time: 04:03 PM, 13 August 2021

Question # 2 of 10 (Start time: 04:04:16 PM, 13 August 2021)

Total Marks: 1

$$Evaluate \quad \int_C f(z)dz \quad where\ C : z(t) = e^{it}\ for\ 0 \leqslant t \leqslant \pi\ and f(z) = 1/z$$

Select the correct option

Reload Math Equations

- $-2\pi i$
- $-\pi i$
- $2\pi i$
- $i\pi$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 04:11:08 PM, 13 August 2021) Total Marks: 1

If C is positively oriented unit circle i.e. {C: |z|=1} then

$$\int_C \frac{e^{2z}}{z^3} dz$$

is

Select the correct option

Reload Math Equations

- $4\pi i$
- $2\pi i$
- $6\pi i$
- $3\pi i$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 04:06:54 PM, 13 August 2021) Total Marks: 1

$$Evaluate\ the\ integral \int_C f(z)dz \quad where\ C : z(t) = e^{it}\ for\ 0 \leqslant t \leqslant \pi\ and f(z) = 1/z^2$$

Select the correct option

Reload Math Equations

- $1 + e^{i\pi}$
- $1 - e^{-i\pi}$ ✓
- $1 - e^{i\pi}$
- $1 + e^{-i\pi}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 11:05:38 AM, 13 August 2021)

If C is positively oriented unit circle i.e. {C: |z|=1} then

$$\int_C \frac{e^{2z}}{z^3} dz$$

is

Select the correct option

○	$3\pi i$
○	$2\pi i$ ✓
○	$6\pi i$
○	$4\pi i$

Question # 3 of 10 (Start time: 10:55:08 AM, 13 August 2021)

Using Cauchy integral formula to compute the value of integral

$$\int_C \frac{z-2}{z+i} dz \text{ where } f(z) = z-2 \text{ and } z_o = -i$$

Select the correct option

○	$2\pi(-1-2i)$ ✓
○	$2\pi(1-2i)$
○	$2\pi(1+2i)$
○	$2\pi(-1+2i)$

$$Evaluate \int_0^{2\pi} \sin(\frac{\pi}{4} + 3e^{i\theta})d\theta \; if \; f(z) = \sin \; z$$

2π

$\sqrt{2\pi}$

$\frac{\pi}{\sqrt{2}}$

$\frac{\pi}{2}$

Click to Save Answe

Question # 7 of 10 (Start time: 04:11:48 PM, 13 August 2021)

Total Marks: 1

$$Consider\ contour\ C: z(t) = e^{it} = Cos\ t\ + i\ Sint\ be\ counter\ clockwise\ where\ -\pi/2 \leqslant t \leqslant \pi/2\ then\ the\ integral$$
$$\int_C \bar{z}(t)dz =$$

Select the correct option

Reload Math Equations

- $-\pi$
- π
- $-i\pi$ ✓
- $i\pi$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 04:07:55 PM, 13 August 2021)

Total Marks: 1

$$Evaluate \int_C f(z)dz \; where \; C : z(t) = (x + iy)\, t \; for \; a \leqslant t \leqslant b \; and \; f(z) = \frac{1}{z}$$

Select the correct option

Reload Math Equations

- $\ln b - \ln a$
- $\frac{(\ln a)}{(\ln b)}$
- $\ln b + \ln a$
- $(\ln a)(\ln b)$

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 04:14:00 PM, 13 August 2021)

Total Marks: 1

If function f is continuous in simple connected domain D and $\int_C f(z)dz$ for every closed contour C in D, then f is analytic in D.

This statement is of:

Select the correct option

Reload Math Equations

- Cauchy Goursat Theorem
- Cauchy theorem
- Morera's Theorem
- Green's Theorem

Click to Save Answer & Move to Next Question

MTH632:Quiz # 3

Quiz Start Time: 04:03 PM, 13 August 2021

Question # 6 of 10 (Start time: 04:11:44 PM, 13 August 2021)

Total Marks: 1

If $f(z)$ is an analytic function in domain D then for $n \geqslant 0$ where n is a positive integer,

Select the correct option

Reload Math Equations

- None of above
- $f^{(n)}(z)$ is analytic in disconnected D
- $f^{(n)}(z)$ is not analytic in simple connected D
- $f^{(n)}(z)$ is an analytic in simple connected D



Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 04:12:53 PM, 13 August 2021) Total Marks: 1

A domain D that is not simply connected is...

Select the correct option

- multiply connected
- doubly connected
- none of above
- triply connected

Click to Save Answer & Move to Next Question

MTH632:Quiz # 3

Quiz Start Time: 04:03 PM, 13 August 2021

Question # 8 of 10 (Start time: 04:13:18 PM, 13 August 2021)

Total Marks: 1

Morera's Theorem is the converse of ...

Select the correct option

- Guass Mean value Theorem
- Cauchy Goursat Theorem
- De Movire's theorem
- Green's Theorem

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 04:06:45 PM, 13 August 2021) T

Let $f(t)$ be continuous function on contour C and L be length of C. For all t in contour C, $|f(t)| \leqslant M$ then

Select the correct option

Reload Ma

- $\left| \int_C f(t)\,dt \right| > ML$
- $\left| \int_C f(t)\,dt \right| < ML$
- $\left| \int_C f(t)\,dt \right| \leqslant ML$
- $\left| \int_C f(t)\,dt \right| \geqslant ML$



Click to Save Answer & Move to Next Que

PM, 13 August 2021)

ion and C and $-C$ be the contours with same geometrical represent

$$\int_C f(-z)dz$$

$$-\int_C f(z)dz$$



$$\int_C f(z)dz$$

$$-\int_{-C} f(z)dz$$

MTH632:Quiz # 3

Quiz Start Time: 04:03 PM, 13 August 2021

Question # 2 of 10 (Start time: 04:04:16 PM, 13 August 2021)

Total Marks: 1

$$Evaluate \int_C f(z)dz \quad where\ C: z(t) = e^{it}\ for\ 0 \leqslant t \leqslant \pi\ and f(z) = 1/z$$

Select the correct option

Reload Math Equations

- $-2\pi i$
- $-\pi i$
- $2\pi i$
- $i\pi$

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 04:03:08 PM, 13 August 2021) Total Marks: 1

A domain D that is not simply connected is...

Select the correct option

- none of above
- doubly connected
- triply connected
- multiply connected



Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 04:03:16 PM, 13 August 2021) Total Marks: 1

$$Let\ \alpha\ lies\ in\ the\ \text{interior}\ of\ contour\ C\ and$$
$$P_n(z) = a_n z^n + a_{n-1} z^{n-1} + a_{n-2} z^{n-2} + \ldots + a_1 z + a_o z^0$$
$$then\ the\ value\ of \int_C \frac{P_n(z)}{z-\alpha} dz\ is\ \ldots\ when\ n = 3$$

Select the correct option

Reload Math Equations

- $2\pi i P_n(\alpha)$
- $2\pi P_n(\alpha)$
- $2\pi P_3(\alpha)$
- $2\pi i P_3(\alpha)$



Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 10:55:08 AM, 13 August 2021)

Using Cauchy integral formula to compute the value of integral

$$\int_C \frac{z-2}{z+i} dz \text{ where } f(z) = z - 2 \text{ and } z_o = -i$$

Select the correct option

○	$2\pi(-1-2i)$ ✓
○	$2\pi(1-2i)$
○	$2\pi(1+2i)$
○	$2\pi(-1+2i)$

MTH632:Quiz # 3

Quiz Start Time: 10:51 AM, 13 August 2021

Question # 2 of 10 (Start time: 10:53:24 AM, 13 August 2021)

Total Marks: 1

If $f(z)$ is complex valued function and C and $-C$ be the contours with same geometrical representation but opposite in direction. Then $\int_{-C} f(z)dz =$

Select the correct option

Reload Math Equations

- $\int_C f(-z)dz$
- $\int_C f(z)dz$
- $-\int_C f(z)dz$
- $-\int_{-C} f(z)dz$

Quiz Is Open Answer & Move to Next Question

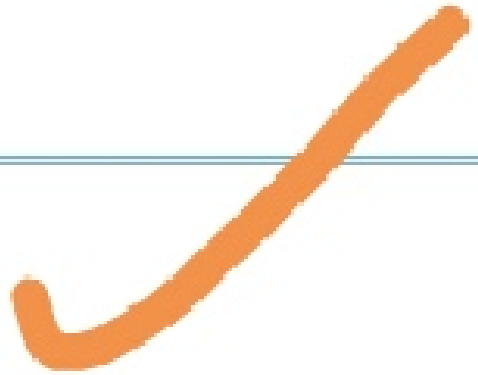
Question # 9 of 10 (Start time: 04:02:31 PM, 13 August 2021)

Total Marks: 1

A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end.

Select the correct option

- False
- True



Click to Save Answer & Move to Next Question

MTH632:Quiz # 3

Question # 1 of 10 (Start time: 10:51:36 AM, 13 August 2021)

If the function f(z) is analytic in domain D and simple connected contour C lies in D then

Select the correct option

- $\int_C f(z)dz = 0$
- $\int_C f(-z)dz = 0$
- $\int_C f(\bar{z})dz = 0$
- $\int_C f(z')dz = 0$

Click

Total Marks: 1

ntours with same geometrical representation but opposite in direction. Then $\int_{-C} f(z)dz =$

Reload Math Equations

$\int_{C} f(-z)dz$

$\int_{C} f(z)dz$

$-\int_{C} f(z)dz$

$-\int_{-C} f(z)dz$

Click to Save Answer & Move to Next Question

Morera's Theorem is the converse of ...

Select the correct option

- Guass Mean value Theorem
- De Movire's theorem
- Green's Theorem
- Cauchy Goursat Theorem ✓

Q 7 of 10 (Start time: 04:39:48 PM, 13 August 2021)

Quiz Start Time:

If function f is an analytic function in simple connected domain where $C_R(z_0) = \{z(t) : |z - z_0| = R\}, 0 \leq t \leq 2\pi$ and R be the radius of circle, then

the correct option

$$f(z_0) = \frac{1}{2\pi i}\int_0^{2\pi} f(z_0 + Re^{it})dt$$

$$f(z_0) = \frac{1}{4\pi}\int_0^{2\pi} f(z_0 + Re^{it})dt$$

$$f(z_0) = \frac{1}{\pi}\int_0^{2\pi} f(z_0 + Re^{it})dt$$

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + Re^{it})dt$$

Type here to search

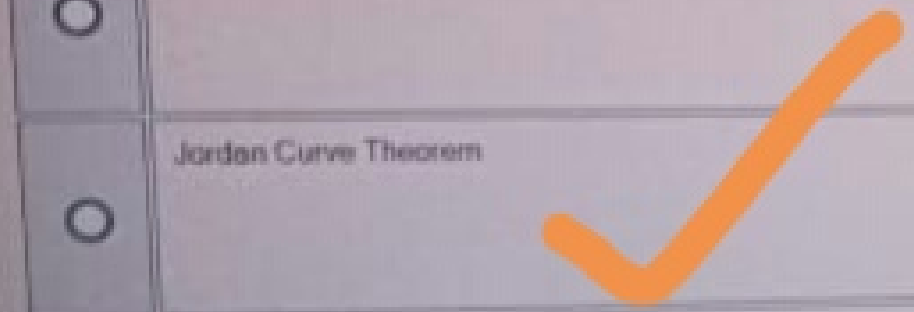
MTH632 Quiz # 3

Question # 5 of 10 (Start time: 04:37:30 PM, 13 August 2021)

Each simple closed curve C divides the plane into interior and exterior domains where one is bounded and other is unbounded. This statement is of ...

Select the correct option

- Cauchy Goursat Theorem
- Jordan Curve Theorem
- Green's Theorem
- Cauchy Theorem



Type here to search

Let z_o and z_1 be two points in simply connected domain D and f be an analytic complex valued function in D and C be contour by joining z_o and z_1 then

$$\int_{z_o}^{z_1} f(s)ds = F(z_o) + F(z_o)$$

$$\int_{z_o}^{z_1} f(s)ds = F(z_1) + F(z_o)$$

$$\int_{z_o}^{z_1} f(s)ds = F(z_o) - F(z_1)$$

$$\int_{z_o}^{z_1} f(s)ds = F(z_1) - F(z_o)$$

If C is simple closed contour with positively oriented such that fixed complex value z_0 lies interior to C, then for integer $n = 1$ we have,

$$\int_C \frac{1}{(z - z_0)^n} dz =$$

Select the correct option

- 0
- πi
- $2\pi i$ ✓
- $-2\pi i$

Evaluate $\int_0^{2\pi} \sin(\frac{\pi}{4} + 3e^{i\theta})d\theta$ *if* $f(z) = \sin\ z$

$\sqrt{2\pi}$

2π

$\frac{\pi}{2}$

$\frac{\pi}{\sqrt{2}}$

C_2 be two simple closed contours with positively oriented such that C_2 is positively oriented as boundary of the region R then

$$\int_{C_1 - C_2} f(z)dz = 0$$

$$\int_{C_1 - C_2} f(z)dz \leqslant 0$$

$$\int_{C_1 - C_2} f(z)dz \geqslant 0$$

$$\int_{C_1 - C_2} f(z)dz \neq 0$$

$\int_C \frac{f(z)}{(z+1)^4} dz$ where $z = e^{2z}$ then this integral is equal to

$\frac{\pi i}{3!} f^{(3)}(1)$

$\frac{\pi i}{3!} f^{(3)}(-1)$

$\frac{2\pi i}{3!} f^{(3)}(1)$

$\frac{2\pi i}{3!} f^{(3)}(-1)$ ✓

Evaluate $\frac{1}{2\pi}\int_0^{2\pi} \exp(\frac{i\pi}{3} + 5e^{i\theta})d\theta$ if $f(z) = \exp(z)$ then we get,

$\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ 

$\cos\frac{\pi}{3} + \sin\frac{\pi}{3}$

$\sin\frac{\pi}{3} + i\cos\frac{\pi}{3}$

$\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}$